

FCC-ee ENERGY CALIBRATION AND POLARIZATION - STATUS AND OUTLOOK

J. Keintzel^{1*}, D.P. Barber², J. Bauche¹, A. Blondel³, A. Bogomyagkov⁴, F. Carlier¹, C. Carli¹, Z. Duan⁵, A. Faus-Golfe⁶, D. Gaskell⁷, K. Hock⁸, M. Hofer¹, W. Höfle¹, G. Iadarola¹, H. Jiang⁶, C. Kiel⁹, I. Koop⁴, A. Korsun⁶, R. Kieffer¹, T. Lefevre¹, A. Martens⁶, S. Mazzoni¹, N. Muchnoi⁴, I. Nikolaev⁴, S. Nikitin⁴, K. Oide³, T. Pieloni⁹, D. Sagan¹⁰, S.M. Saenz⁶, D. Sittard¹, J. Tamazirt⁶, R. Tomás¹, E. Torrence¹¹, L. van Riesen-Haupt⁹, J. Wenninger¹, G. Wilkinson¹, Y. Wu⁹, Z. Zhang⁶

¹CERN, Geneva, Switzerland; ²DESY, Hamburg, Germany

³DPNC, University of Geneva, Switzerland

⁴Affiliated with an institute formerly covered by a cooperation agreement with CERN

⁵IHEP, Beijing, China; ⁶IJCLab, Orsay, France; ⁷Jefferson Lab, Newport News, VA, USA

⁸BNL, Upton, USA; ⁹EPFL, Lausanne, Switzerland

¹⁰Cornell University, NY, USA; ¹¹University of Oregon, Eugene, USA

Abstract

The FCC-ee aims at high-precision particle physics experiments with beam energies from 45.6 to 182.5 GeV, corresponding to the Z-pole up to above the top-pair-threshold. These goals demand, among others, a precise knowledge of the center-of-mass energy and, hence, the beam energies. By depolarizing previously polarized pilot bunches up to 80 GeV and recording the change of polarisation with a 3D polarimeter, it is aimed to determine the spin tune and thereby achieve a systematic uncertainty on the beam energy in the order of tens of keV. The latest progress on the energy calibration and polarisation working group is reported here.

INTRODUCTION

Within the framework of the European Strategy Update for particle physics, the Future Circular electron-positron Collider (FCC-ee) is being proposed as a potential new flagship collider at CERN, followed by a hadron machine (FCC-hh) [1]. The FCC-ee is designed to perform high precision physics experiments with beam energies from 45.6 GeV to 182.5 GeV, corresponding to measurements at the Z-pole up to above the $t\bar{t}$ -threshold. Recently, the FCC Feasibility Report has been published in 3 volumes, covering physics and experiments [2], accelerator design and technology [3] and civil engineering and environmental studies [4].

To probe a wide range of electroweak observables, whose overall consistency can then be assessed, the knowledge of the collision energy is one of the key inputs to those measurements. Beam energy is measured by using the technique of Resonant Depolarisation (RDP), where previously polarized low-intensity pilot bunches (10^{10} particles per bunch (ppb)) are depolarized with RF-kickers and the change of polarisation is recorded using Compton polarimeters, which is feasible until about 80 GeV beam energy. These measurements are foreseen to take place several times per hour, while nominal-intensity physics bunches ($\approx 2 \times 10^{11}$ ppb) are brought to collision. As a complementary technique

to RDP Free Spin Precession (FSP) could be performed, where the spin is rotated into the horizontal plane, and the precession frequency can then be measured directly. The target absolute precision of the energy measurement at the Z-pole from RDP or FSP is about 100 keV. Various expected uncertainties are summarized in Table 1 [2]. The collision energy is obtained by measurements of the beam energy for both beams including corrections specific to each Interaction Point (IP).

Physics measurements exactly at the H-resonance, corresponding to beam energies of 62.5 GeV, require additionally a small centre-of-mass energy ($\sqrt{s}$) spread, which is achieved using monochromatisation. One method of reducing the collision energy spread is by introducing dispersion which demands lattice modifications.

The Energy Calibration, Polarisation and Monochromatisation (EPOL) team is studying challenges related to $\sqrt{s}$ determination and monochromatisation options, where a brief status report, similar to [5], highlighting recent progress, is given in the following sections. An extended report can be found in [2, 3, 6].

Table 1: $\sqrt{s}$ -related Uncertainties from Ref. [2]

Uncertainty	m_Z (keV)	Γ_Z (keV)	m_W (keV)
Absolute	100	2.5	150
Point-to-point	14	11	50
Sample size	1	1	3
Energy spread	–	5	–
Total $\sqrt{s}$ -related	101	12	158
FCC-ee statistical	4	4	180

BEAM POLARISATION

The polarisation of electron and positron beams builds up naturally over time thanks to the Sokolov-Ternov effect [7]. To first order, the spin tune is directly proportional to the average beam energy by $\nu_0 = a\gamma_{\text{rel}}$, with the gyro-magnetic

* jacqueline.keintzel@cern.ch

anomaly a and the Lorentz-factor γ_{rel} . In an error-free flat machine the polarisation is fully vertical, and hence the invariant spin axis is $\vec{n}_0 \parallel \hat{y}$. Studies using the Global Hybrid Correction (GHC) optics [8], the maximum theoretical polarisation is above 90 %, which is, however, diminished, by solenoids, misalignment errors or field errors. These errors could also introduce a spin tune shift, and hence, the measured spin tune would no longer correspond to the beam energy. Tuning studies including spin dynamics [9], as also reported in [10] show that assuming up to 100 μm arc and 25 μm Interaction Region (IR) rms misalignment errors at least 10 % polarisation is achieved with an absolute energy shift below the target. A preliminary systematic study [11] shows that for 50 μm alignment tolerance, the relation between the spin tune and the beam energy is accurate to within fluctuations of a few keV. Future studies should be performed with more realistic misalignment errors along with various magnetic field imperfections [10]. These studies should also evaluate the performance of the Local Chromaticity Correction (LCC) [12] optics and perform detailed comparison studies [13].

Two solenoid compensation schemes are investigated for the FCC-ee, namely placing the compensation solenoid between the IP and the final focus (local scheme), or placing it afterwards (non-local scheme). The latter leads to interleaved spin deflections around the longitudinal and the horizontal axis, resulting in a rotation from the vertical direction of $\vec{n}_0$ estimated to roughly 10 μrad . This deflection reduces the asymptotic polarisation, simulated in SAD, to about 1 %. Current studies aim at restoring the polarisation by dedicated orbit bumps, as performed in LEP.

Spin dynamics is being implemented in the simulation framework Xsuite [14]. First studies focus on implementing spin tracking through elements.

DEPOLARISATION

Requirements for the depolarizer are based on the RDP method. The depolarizer is a selective, electromagnetic kicker using transverse fields (RF-kicker), with a TEM-wave travelling towards the beams.

For RDP varying excitation frequency is applied to one pilot bunch at a time. Once the driving frequency is equal to the spin-tune, the polarisation vector is rotated away from the vertical direction, leading to depolarisation or spin flip. We note that RDP is only feasible if the spin modulation index fulfils $\nu_0 \sigma_E / Q_s \lesssim 1.2 - 1.4$, with the energy spread σ_E and the synchrotron tune Q_s , which ensures a low number of synchrotron-tune sidebands inside the distribution of the spin tune. RDP scans have been performed at numerous facilities, including recent studies at KARA [15]. Since a single kick would lead to a propagating orbit through the machine, a closed-bump must be applied. In order to achieve a spin rotation, dipoles must be located within the closed orbit bump. It is found that for the GHC optics the regular arc cells would allow sufficient rotation over four FODO cells. Numerous studies are currently ongoing to determine the

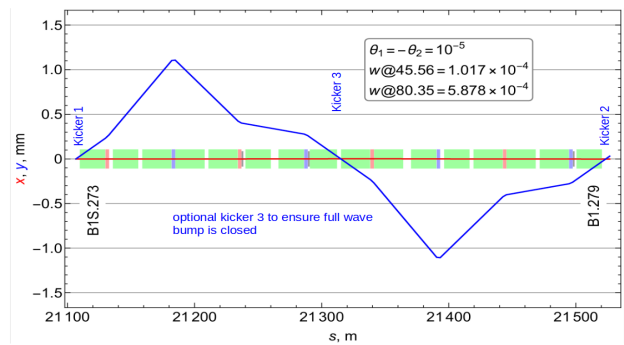


Figure 1: Proposed closed-orbit bump in the arcs to depolarize pilot bunches [18].

most suited parameters for the RDP scan [16, 17]. Studies propose applying one closed orbit bump of 10 μrad placed in the arcs, as shown in Fig. 1 [18]. Since this would lead to a vertical peak orbit above 1 mm, it is proposed to distribute it over four closed-orbit bumps per beam, equally distributed over the machine [19]. For a closed-orbit at least two kickers per bump are required, constraining the vertical phase advance to 180°. A weaker third kicker would ease this constraint. The third correction kicker could be designed as a slightly shorter strip-line with less RF power installed, as it only has to provide corrections to the bump. Hence, a total of 16 to 24 kickers are required. Assuming that each kicker provides 2.5 μrad , the current design features a strip-line design of 1 m length with four electrodes, operating around 40 MHz. Combining the depolarizer with the transverse feedback system [20], or with AC-dipoles for optics measurements could also be investigated.

Keeping the nominal vacuum chamber diameter of 70 mm requires an RF power per kicker port (electrode) of 35 kW, and it is therefore suggested to reduce the vacuum chamber at each kicker to 26 mm with electrodes at a distance of 9 mm from the beam. This reduces the RF power per kick to 9.04 kW. Impedance calculations remain to be performed. In Table 2 the configuration at PA is given for both beams. A similar system is installed close to the other experimental straight sections.

As a complementary technique FSP [29] is being investigated, where the spin is flipped into the horizontal plane, and the coherent (free-spin) precession is then observed, ideally turn-by-turn. The spin tune is then retrieved by a Fourier transform, which also yields the full spin spectrum of the spin motion. This technique would require a kicker pulse about ten times stronger than that planned for the RDP measurement.

POLARIMETER

Measurements of the polarisation of the FCC-ee beams will be performed through the process of inverse Compton scattering, where a 515 nm laser beam crosses the particle beam under an angle of 8°. Measuring back-scattered electrons or positrons with the minimum energy and Compton photons allows real-time measurement of the beam energy, which can attain a statistical precision of 10^{-3} per second.

Table 2: Specifications of Kickers at Point PA [3]

Location	Function	Length	Power
left of IP e^-	open bump	1.0 m	9.04 kW
left of IP e^-	correction	0.75 m	4.5 kW
left of IP e^-	close bump	1.0 m	9.04 kW
right of IP e^+	open bump	1.0 m	9.04 kW
right of IP e^+	correction	0.75 m	4.5 kW
right of IP e^+	close bump	1.0 m	9.04 kW

The polarimeter is schematically shown in Fig. 2. The most suitable location for the polarimeter using the GHC optics is in the straight section 830 m upstream of the experiment IPs. The dispersion suppression dipole is then also used as the system spectrometer magnet. This location is followed by 100 m of field-free propagation, allowing sufficient separation of the Compton products from the main beam. The room that houses the laser would be installed in a shielded region as close as possible to the laser interaction point. In the current baseline the laser room is not accessible during machine operation, and therefore, two polarimeter systems per beam are foreseen. Furthermore, the stringent physics goals require an availability for polarimeter measurements of 95 %. Thus, it will be necessary to thoroughly investigate and validate the reliability of a system based on fully remote laser control in future studies. A detailed description is given in [28].

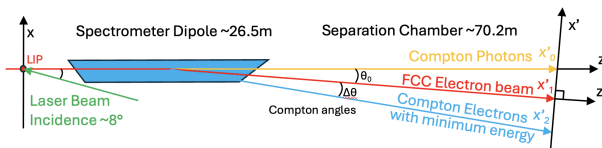


Figure 2: Schematic drawing of the FCC-ee polarimeter [28].

OPERATION

In the current baseline 160 low-intensity pilot bunches are injected at the start-of-fill and polarised using asymmetric polarisation wigglers [6]. Once 10 % polarisation is achieved, wigglers are switched off, and all the nominal-intensity bunches are injected and brought to collision. Assuming a pilot bunch lifetime of 20 h [23], every 7.5 min one of those bunches could be used to determine the beam energy; and by the time the last pilot bunch has been depolarised for the first time, the pilot bunches that were depolarised first will have naturally reacquired sufficient polarisation to be measured again. Availability studies [21] suggest that injecting pre-polarised pilot bunches could significantly enhance the time available for physics, especially when considering failure scenarios. Preliminary studies [22] have found no show-stopper, yet a careful evaluation of the injector design, which must be suitable for generating polarised electrons and positrons and allow to keep sufficient

polarization through the full injector chain and through the energy ramp, remains to be performed.

IP-SPECIFIC CORRECTIONS

Both RDP and FSP measurements determine the average beam energy over the ring, which itself depends on numerous mechanisms including synchrotron radiation losses or resistive wall impedance losses. Furthermore, the $\sqrt{s}$ needs to be determined at each IP individually, and is, additionally, modified by e.g. beamstrahlung losses or systematic offsets between pilot and colliding bunches. More details are reported in [23]. One crucial parameter that must be controlled within a view percent of the beam size are the collision offsets, demanding an extremely precise IP feedback system [24, 25].

MONOCHROMATISATION

Measuring the Yukawa couplings is foreseen at a special beam energy run of 62.5 GeV beam energy. Since this resonance has a width of only 4.1 MeV, the $\sqrt{s}$ spread from initially 50 MeV at that energy must be reduced. Monochromatization is achieved with a chicane before the IP introduces opposite horizontal or vertical dispersion of, respectively 0.105 m and 1 mm. This demands also severe modification of the lattice and optics in the IR. More information is given in [26, 27].

SUMMARY AND OUTLOOK

The status of the FCC-ee EPOL working group has been detailed in the FSR [2, 3] and in an extended report [6], with only a brief summary has been described here. Although a baseline scheme for calibration of the collision energy and monochromatisation options have been developed, these studies must be refined and alternative schemes should be considered to further improve performance and operational. It is important to have a reliable procedure to accurately translate the mean beam energy, to the local collision energy at all IPs. Full simulations of this procedure still have to be conducted, incorporating various contributions. Attention must also be paid to the control of energy shifts from possible dispersion effects at each IP. Furthermore, any effects that might bias the assumed proportionality between the spin tune and the mean beam energy will need to be investigated. Detailed technical designs will now be made of the polarimeter and depolariser systems and the feasibility of injecting pre-polarized bunches will be investigated. Further investigations will also take place regarding the feasibility of the electron Yukawa measurement. In particular, new and refined schemes will be investigated with the aim of improving the monochromatisation of the collision energy, including a procedure to monitor and adjust the collision energy in real time.

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